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LETTERS TO THE EDITOR

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COMMUNICATIONS

Concerning the applicability of density functional methods to atomic and molecular negative ions

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There is concern within theoretical chemistry that density functional methods are fundamentally in error for negative ions. We have tested this hypothesis by evaluating electron affinities for F and F_2 with a variety of density functionals and extremely large, diffuse basis sets. In addition, we have observed the behavior of a known unbound system Ne^-. We have found no convincing evidence to support the claims of negative ion instability with density functional methods. © *1996 American Institute of Physics.* [S0021-9606(96)03126-1]

There are a number of statements in the literature indicating that density functional methods are inappropriate for negative atomic and molecular ions.[1–3] In particular, local density functional methods suffer from a ''spurious interaction of each electron with itself.''[4] As a result, at large distances from the nuclear center, the potential for an N-electron ion goes to $-(Z-N)/r$ rather than the correct limit $-(Z-N+1)/r$.[5,6]

In very recent studies,[7] we have found that density functional methods provide reasonable predictions for the family of fluoride anions AF_n^-, where A=P, S, and Cl, and n=1–7. Thus the literature statements concerning the inapplicability of density functional schemes have been puzzling to us. From some comments on our work,[7] others were puzzled as well. For example, the referee of one of these papers stated, ''Suppose a complete basis set was used. For the negative ions, that would result in a departing plane wave function for the odd electron of highest energy, since its orbital energy is likely positive.''

In this Communication we test the above hypothesis for two simple cases, the fluorine atom and the fluorine molecule. The electron affinities of both F and F_2 (3.40 and 3.01 eV, respectively[8,9]) are accurately determined experimental values to which theoretical predictions can be compared. In addition, F^- is specifically mentioned in connection with the above hypothesis,[3] and thus well suited to test the notion that density functional methods necessarily fail for anions.

A critical ingredient in the test outlined above is a systematic procedure for expansion of the basis set. For this purpose we initially chose Dunning's augmented correlation consistent basis sets.[10] For a variety of properties, these new basis sets appear to provide a steady path to convergence with respect to the complete basis set limits.

The density functionals chosen here include the same four methods used in our studies of AF_n^- molecules.[7] The first (BLYP) uses Becke's 1988 exchange functional (B)[11] with Lee, Yang, and Parr's correlation function (LYP).[12] The B3LYP method is a HF/DFT hybrid method using Becke's three-parameter semiempirical exchange functional (B3)[13] with the LYP correlation functional. Another HF/DFT hybrid method, BHLYP, was formed from a modified version of Becke's half-and-half exchange functional (BH)[14] along with the LYP correlation functional. Finally, the BP86 method was comprised of the exchange and correlation corrections of Becke (B) and Perdew (P86),[15] respectively. In addition to the four DFT methods used in Ref. 7, we have also applied the less sophisticated local-spin-density exchange-correlation approximation (LSDA) to F, F^-, F_2, and F_2^-. All computations were carried out with the GAUSSIAN 94 program.[16]

The present results are summarized in Table I. Note that for F_2 and F_2^-, the equilibrium bond distances were predicted separately at each level of theory.

The principle conclusion from the present research is that the DFT electron affinities are rather insensitive to basis set, after one reaches the augmented correlation consistent polarized valence double ζ (aug-cc-pVDZ) basis set. For the fluorine atom, the range of electron affinities (as a function of basis set) is 0.017 eV (BLYP), 0.035 eV (B3LYP), 0.060 eV (BHLYP), 0.026 eV (BP86), and 0.013 eV (LSDA). For the fluorine molecule, the range of electron affinities is 0.103 eV (BLYP), 0.147 eV (B3LYP), 0.209 eV (BHLYP), 0.136 eV (BP86), and 0.144 eV (LSDA). Thus there is no evidence that the last electron in either F^- or F_2^- is trying to remove itself from the anion.

A final test was provided by placing a single diffuse s-type function (orbital exponent α_s=0.0986) 10 Å away from the F atom in F^-. In the case of F_2 and F_2^-, the additional function was placed perpendicular to the F–F bond and 10 Å away from the midpoint. The electron affinities

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TABLE I. The electron affinities of F and F_2 with various density functionals as a function of basis set size.[a,b]

	aug-cc-pVDZ	aug-cc-pVTZ	aug-cc-pVQZ	aug-cc-pV5Z
BLYP				
F	3.692	3.672	3.675	3.681
F_2	3.740	3.655	3.642	3.637
B3LYP				
F	3.561	3.530	3.526	3.527
F_2	3.790	3.670	3.649	3.643
BHLYP				
F	2.960	2.921	2.902	2.900
F_2	3.591	3.415	3.391	3.382
BP86				
F	3.782	3.759	3.756	3.759
F_2	3.697	3.591	3.569	3.561
LSDA				
F	4.643	4.631	4.630	4.633
F_2	3.862	3.753	3.724	3.718

[a]All electron affinities are reported in eV.
[b]Experimental electron affinity of F=3.40 eV; Ref. 8. Experimental electron affinity of F_2=3.01 eV; Ref. 9.

computed in this manner are the same as those reported in Table I.

In order to address the question of positive orbital energies, the highest occupied orbital energies are reported in Table II. For F_2^- the highest occupied molecular orbital (HOMO) energy is *negative* for all basis sets and all methods. For F^- the highest occupied orbital energies are indeed positive with the pure (no admixture of Hartree–Fock exchange) density functionals BLYP, BP86, and LSDA. For the hybrid functional BHLYP, this orbital energy is negative and for the B3LYP hybrid functional it is nearly zero and becomes slightly negative with the aug-cc-pV5Z basis set. While a positive HOMO energy in Hartree–Fock theory would indicate an unbound electron via Koopmans' theorem, the physical significance of the orbitals obtained from density functional methods is questionable. Although it has been shown[17] for exact solutions of the exchange correlation potential, V_{xc}, that Koopmans' theorem is applicable to only the highest occupied molecular orbital (HOMO), the effect of this result with approximate V_{xc} is unclear.

Focusing on the LSDA, which is most commonly associated with the problem of negative ion instability in density functional methods, it is apparent that the F^- highest occupied orbital energy is decreasing, though very slightly, as the size of the basis set increases. Therefore, extra even tempered[18] p-type diffuse functions (α_p=0.019 97, 0.007 12, 0.002 56, 0.000 92, 0.000 33, 0.000 12, 0.000 04, 0.000 02) were added to the aug-cc-pV5Z basis set. A total of eight extra, *extremely* diffuse basis functions were added sequentially in order to dispel any thoughts of an insufficiently diffuse basis set. As these diffuse functions were added, the highest occupied orbital energy of F^- continued to gradually decrease, until increasing slightly and leveling out to a value of +0.019 Hartree. The electron affinity of F increased with the addition of diffuse functions until plateauing at a value of 4.695 eV, only 0.062 eV above the aug-cc-pV5Z result.

TABLE II. Highest occupied orbital energies[a] for F, F^-, F_2, and F_2^-.

	aug-cc-pVDZ	aug-cc-pVTZ	aug-cc-pVQZ	aug-cc-pV5Z
BLYP				
F	−0.378	−0.380	−0.380	−0.380
F^-	0.059	0.057	0.055	0.053
F_2	−0.352	−0.351	−0.351	−0.351
F_2^-	−0.020	−0.020	−0.021	−0.021
B3LYP				
F	−0.451	−0.452	−0.452	−0.452
F^-	0.003	0.002	0.001	−0.000
F_2	−0.423	−0.421	−0.421	−0.421
F_2^-	−0.078	−0.078	−0.078	−0.079
BHLYP				
F	−0.547	−0.547	−0.547	−0.547
F^-	−0.071	−0.072	−0.072	−0.072
F_2	−0.517	−0.514	−0.514	−0.514
F_2^-	−0.119	−0.117	−0.117	−0.117
BP86				
F	−0.383	−0.384	−0.383	−0.383
F^-	0.055	0.053	0.051	0.049
F_2	−0.354	−0.353	−0.352	−0.352
F_2^-	−0.022	−0.022	−0.023	−0.023
LSDA				
F	−0.399	−0.400	−0.400	−0.400
F^-	0.035	0.034	0.032	0.031
F_2	−0.369	−0.369	−0.368	−0.368
F_2^-	−0.034	−0.033	−0.034	−0.034

[a]All orbital energies are reported in Hartree.

Finally, in order to model the behavior of a known unbound system, the electron affinity of Ne was determined with LSDA (Table III). Ne is a noble gas, and therefore has no electron affinity in the conventional sense. The finite basis sets used in this study, however, could conceivably impose an artificial barrier on this extra electron leading to a bound solution. Ne^- lies higher in energy than Ne and therefore the electron affinity is negative. As the basis set is expanded, the electron affinity becomes less negative until reaching a value less than 0.001 eV when two even-tempered diffuse s-type functions (α_s=0.035, 0.012) are added to the aug-cc-pV5Z basis set. With all basis sets, the basis function with the largest contribution to the highest occupied orbital was the most diffuse s-type function. This is in contrast to F^- where the diffuse functions contributed a negligible amount. In addition, when a single diffuse s-type function was placed 10 Å

TABLE III. Electron affinity of Ne and highest occupied orbital energies of Ne^- with LSDA.[a,b]

	aug-cc-pVDZ	aug-cc-pVTZ	aug-cc-pVQZ	aug-cc-pV5Z	aug-cc-pV5Z$^+$	aug-cc-pV5Z^{++}
EA	−6.1965	−5.144	−4.415	−3.713	−0.784	0.000
ϵ	0.350	0.303	0.268	0.235	0.087	0.034

[a]All electron affinities are reported in eV.
[b]Highest occupied orbital energies (ϵ) of Ne^- are reported in Hartree.

away from the Ne center, this basis function was essentially the sole contributor to the highest occupied orbital.

We conclude that density functional schemes are applicable to negative ions in which the extra electron is bound as strongly as in F^-. It is unclear whether the ''spurious self-interaction'' alluded to in the literature[4–6] will become significant for loosely bound negative ions. Of course, this does not mean that DFT methods are a panacea for the prediction of electron affinities (EA). The best agreement with experiment for EA(F) is provided by the B3LYP method, for which the theoretical electron affinity lies 0.13 eV above experiment. For F_2, however, the BHLYP method does the best, yielding an EA value 0.37 eV higher than experiment.

While there is a formal basis for the literature statements[1–6] that local density functional methods fail to yield bound negative ions, there are still important questions which remain unanswered. Specifically, where does Koopmans' theorem break down for the HOMO in DFT, do these arguments apply to atoms only or to molecular anions as well, and are they applicable also to nonlocal exchange-correlation functionals? The empirical evidence certainly suggests that these methods can be useful, particularly when calibration via a few experimental electron affinities is possible.[7] As Table I clearly shows, the three pure density functional methods employed here overestimate the correlation effects, providing an upper bound for electron affinities. On the other hand, conventional methods tend to underestimate correlation effects leading to a lower bound for electron affinities. When both methods are used in conjunction, error bars may be determined to accurately pinpoint electron affinities.[19]

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